

Klebsiella Pneumonia — Comprehensive Disease Characteristics Report

Disease: Klebsiella pneumonia (infection caused by *Klebsiella pneumoniae*) **MONDO ID:** MONDO:0030602 · **Pathogen taxonomy:** NCBI:txid573 · **Category:** Infectious Disease **Report type:** Multi-iteration autonomous literature synthesis (10 confirmed findings; 46 papers reviewed)

Summary

Klebsiella pneumonia is an acute bacterial infection caused by the encapsulated, non-motile, Gram-negative rod *Klebsiella pneumoniae*, a member of the *Enterobacteriaceae*. Clinically the disease is not a single entity but a spectrum: lobar/nosocomial pneumonia, ventilator-associated pneumonia, bloodstream infection (bacteremia/sepsis), urinary tract infection, and — for a distinct **hypervirulent** pathotype — community-acquired pyogenic liver abscess with metastatic spread. *K. pneumoniae* is one of the **ESKAPE** organisms and a **WHO critical-priority** antimicrobial-resistance (AMR) pathogen, making it a central actor in the global AMR crisis [PMID: 42178970](#). Unlike the Mendelian disorders this template was designed around, Klebsiella pneumonia is an **infectious, non-heritable disease**: its "causal genes" are bacterial virulence and resistance genes, not human germline variants, and host genetic susceptibility loci are minor relative to healthcare exposures.

The investigation converged on a **two-pathotype model**. **Classical *K. pneumoniae* (cKp)** is an opportunistic nosocomial pathogen that infects debilitated, catheterized, ventilated, or antibiotic-exposed patients and is the leading cause of neonatal sepsis in low- and middle-income countries (LMICs); much of its burden is driven by in-hospital transmission [PMID: 42127138](#). Its danger lies in **carbapenem resistance** mediated by KPC, NDM, and OXA-48 carbapenemases, which pushes in-hospital mortality of bloodstream infection to 30–50% [PMID: 42734011](#), [PMID: 42518856](#). **Hypervirulent *K. pneumoniae* (hvKp)** infects younger, immunocompetent hosts in the community, causing invasive liver abscess through K1/K2 capsule serotypes, *rmpADC*-driven hypermucoviscosity, and multiple siderophore systems [PMID: 42021129](#). The two pathotypes are converging on mobile plasmids (e.g., ST11 KPC-2 hvKp), an emerging worst-case threat that fuses resistance with virulence [PMID: 32576652](#).

Mechanistically, pathogenesis runs from **gut/mucosal colonization** → **barrier breach** → **capsule- and siderophore-mediated immune evasion** → **TLR4- and NLRC4-inflammasome-driven neutrophil/macrophage recruitment** → **either clearance or, when immunity fails or the organism is resistant, invasive infection, sepsis, and death** [PMID: 22547706](#), [PMID: 36018281](#). Diagnosis rests on culture with MALDI-TOF species identification plus molecular/phenotypic resistance detection; treatment is susceptibility-guided, favoring newer β -lactam/ β -lactamase-inhibitor combinations (ceftazidime-avibactam, meropenem-vaborbactam, imipenem-relebactam) and cefiderocol for metallo- β -lactamase producers; and prevention hinges on antibiotic stewardship and infection control, because no vaccine is yet licensed.

1. Disease Information

Overview. *Klebsiella pneumoniae* infection by *Klebsiella pneumoniae*, historically described by Carl Friedländer (hence "Friedländer's bacillus" / "Friedländer's pneumonia") as a cause of severe lobar pneumonia classically producing thick, blood-tinged "currant-jelly" sputum. Modern disease is dominated by **healthcare-associated** presentations — hospital-acquired and ventilator-associated pneumonia, catheter-associated UTI, and central-line bloodstream infection — plus **community-acquired invasive syndromes** driven by the hypervirulent pathotype.

Key identifiers. - **MONDO:** MONDO:0030602 (*Klebsiella pneumoniae*) - **Pathogen (NCBI Taxonomy):** *Klebsiella pneumoniae* txid573 - **MeSH:** *Klebsiella Infections* (D007710); *Pneumonia, Bacterial* - **ICD-10:** J15.0 (Pneumonia due to *Klebsiella pneumoniae*); A49.8/B96.1 (*Klebsiella* as cause of disease classified elsewhere) - **ICD-11:** CA40 (bacterial pneumonia) with pathogen extension; infection codes under 1B/1C - **OMIM / Orphanet:** Not applicable — this is an acquired infectious disease, not a Mendelian disorder. There is no OMIM phenotype entry; Orphanet does not list it as a rare genetic disease.

Synonyms / alternative names. Friedländer's pneumonia; Friedländer's bacillus infection; *Klebsiella pneumoniae* infection; *Klebsiella* infection (when generalized); (for the invasive pathotype) hypervirulent *Klebsiella pneumoniae* infection / invasive *Klebsiella* syndrome.

Data source type. The evidence base is overwhelmingly **aggregated disease-level** — surveillance cohorts, genomic epidemiology, meta-analyses, and clinical guidelines — supplemented by individual case reports/series for the hypervirulent syndrome. It is not derived from a single-patient EHR data file.

2. Etiology

Primary cause — infectious. The disease is caused by infection with *K. pneumoniae*. There is **no human germline genetic cause**. Two ecological/virulence pathotypes drive distinct etiologies:

Feature	Classical (cKp)	Hypervirulent (hvKp)
Typical host	Debilitated, hospitalized, immunocompromised	Younger, often immunocompetent
Setting	Nosocomial	Community-acquired
Hallmark syndrome	HAP/VAP, BSI, UTI, neonatal sepsis	Pyogenic liver abscess, metastatic (endophthalmitis, meningitis)
Key genetics	Carbapenemases (KPC/NDM/OXA-48)	K1/K2 capsule, <i>rmpADC</i> , aerobactin/salmochelin/yersiniabactin
Mortality driver	Antibiotic resistance	Dissemination/metastasis

Risk factors (host/environmental — quantified). A meta-analysis of 30 studies (5,075 cases) established that carbapenem-resistant *K. pneumoniae* (CRKP) infection is driven by **healthcare exposures and prior antibiotics, not host demographics** [PMID: 31525540](#):

Risk factor	Odds ratio (95% CI)
Carbapenem exposure	3.99 (2.86–5.56)
ICU admission	3.25 (2.36–4.47)
Glycopeptide exposure	3.08 (1.93–4.91)
Central venous catheterization	2.93 (2.00–4.28)
Mechanical ventilation	2.91 (1.96–4.31)
Indwelling (urinary) catheter	2.62 (1.65–4.17)
Any prior antibiotic exposure	2.53 (1.56–4.11)
Nasogastric intubation	2.38 (1.22–4.62)
β -lactam/ β -lactamase-inhibitor exposure	2.28 (1.37–3.80)
Quinolone exposure	1.75 (1.38–2.22)
Surgery	1.59 (1.08–2.34)
Immunosuppression	1.47 (1.14–1.90)

Notably, **age, sex, and diabetes mellitus were NOT associated** with CRKP infection in this meta-analysis (a useful negative finding) — although diabetes is a well-recognized risk factor specifically for **hvKp liver abscess** in the Asian literature. Additional exposure-based risks include prolonged mechanical ventilation and prolonged ECMO support, where prior antibiotic use before ECMO (RR 1.25) and longer ventilation (RR 1.31) independently predicted healthcare-associated infection [PMID: 42685911](#).

Genetic risk factors (human). No robust, replicated human susceptibility loci for *K. pneumoniae* disease were identified in this investigation. Innate-immune pathway integrity (TLR4, NLRC4/inflammasome, IL-1 β signaling) is mechanistically protective in models, implying that host defects in these pathways would increase susceptibility, but this remains inferred rather than demonstrated in human genetic association studies.

Protective factors. Intact mucosal barriers, a diverse gut microbiome that resists colonization by multidrug-resistant Enterobacterales, and competent innate immunity are protective. Gut microbiome composition modulates intestinal colonization by MDR Enterobacterales [PMID: 41009869](#), [PMID: 38659243](#) — implying microbiome preservation (avoiding unnecessary broad-spectrum antibiotics) is protective. No validated protective human genetic variant was found.

Gene–environment interaction. The dominant interaction is **antibiotic exposure × bacterial resistance genotype**: antibiotic pressure selects for carbapenemase-carrying clones in the gut reservoir, which then cause endogenous infection or spread nosocomially. This is an environment-(bacterial)genotype interaction rather than a human GxE effect.

3. Phenotypes

Phenotypes are clinical manifestations of infection, varying by syndrome. Onset is typically **acute**; severity ranges **moderate to severe**; frequency figures are syndrome-dependent.

Phenotype	Type	Syndrome	Suggested HPO term
Pneumonia / productive cough	Clinical sign	HAP/VAP, CAP	HP:0002090 (Pneumonia); HP:0031246 (Productive cough)
Fever	Symptom	All	HP:0001945 (Fever)
Dyspnea	Symptom	Pneumonia, sepsis	HP:0002094 (Dyspnea)
"Currant-jelly" / bloody sputum	Clinical sign	Friedländer pneumonia	HP:0002105 (Hemoptysis)
Sepsis / septic shock	Clinical sign	BSI	HP:0100806 (Sepsis)
Bacteremia	Laboratory	BSI	HP:0031864 (Bacteremia)*
Liver abscess	Physical/ imaging	hvKp	HP:0100523 (Hepatic abscess)*
Leukocytosis	Laboratory	All	HP:0001974 (Leukocytosis)
Elevated D-dimer / coagulopathy	Laboratory	Severe BSI	HP:0003581-adjacent (coagulation abnormality)
Urinary tract infection	Clinical sign	Catheter-associated	HP:0000010 (Recurrent UTI)
Neonatal sepsis	Clinical sign	LMIC neonates	HP:0001945 + neonatal onset
Metastatic endophthalmitis / meningitis	Clinical sign	hvKp dissemination	HP:0000623 / HP:0001287

*Term suggestions; verify exact HPO ID before ingestion.

Age of onset. Bimodal — **neonatal** (LMIC sepsis) and **older adults** (nosocomial; mean age $\approx$ 66 years, male predominance $\approx$ 70% in BSI cohorts). hvKp liver abscess tends to occur in middle-aged adults.

Quality-of-life impact. Severe infection causes ICU-level morbidity, prolonged hospitalization, and long-term functional impairment in survivors of sepsis; disease-specific QoL instruments were not identified in the literature reviewed.

4. Genetic / Molecular Information (bacterial)

Because this is an infectious disease, the relevant genetics are the **pathogen's virulence and resistance determinants**, not human variants.

Resistance genes (carbapenemases). Carbapenem resistance is mediated chiefly by three carbapenemase families [PMID: 42734011](#), [PMID: 42439974](#): - **KPC** (*blaKPC*; class A) — dominant in many regions; 88.9% (48/54) of CRKP BSI isolates in a Chinese cohort. - **NDM** (*blaNDM*; class B metallo-β-lactamase) — 7.4% in the same cohort; alters therapy because MBLs hydrolyze most β-lactams and are not inhibited by avibactam/vaborbactam. - **OXA-48** (*blaOXA-48*; class D) — predominant (50.4%) in a southern Saudi Arabia CRE cohort, illustrating strong geographic variation. Additional/associated genes: *blaVIM* (MBL), *blaSHV* and *blaCTX-M* (ESBLs).

Virulence genes. In CRKP BSI, virulence genes *ycfM*, *wabG*, and *entB* were present in **all** strains; *fimH* 98.15%, *mrkD* 96.30%, *uge* 92.60%, *kpn* 92.60% [PMID: 42734011](#). For **hvKp**, the defining loci are the **capsular polysaccharide serotypes K1/K2**, the **regulator of mucoid phenotype** operon *rmpADC* (drives hypermucoviscosity), and multiple **siderophore** systems — **aerobactin**, **salmochelins** (*iroBCDN*), and **yersiniabactin** (*ybt*) — frequently carried on large virulence plasmids. A characterized ST111 hvKp isolate harbored a **181 kb IncHI1B/IncFIB virulence plasmid** encoding *rmpADC* and salmochelins (*iroBCDN*) plus yersiniabactin; CRISPR/Cas9 plasmid curing followed by murine infection **confirmed the plasmid's causal contribution** to hypervirulence [PMID: 42021129](#).

Lineages (multilocus sequence types). hvKp liver abscess is canonically linked to **ST23**, **ST65**, **ST86**; convergent resistance-plus-virulence clones include **ST11** (KPC-2-producing hvKp).

Epigenetics / chromosomal abnormalities / human modifier genes. Not applicable to this infectious disease. (Bacterial DNA methylation/restriction-modification systems exist but are not disease-defining here.)

5. Environmental Information

Infectious agent. *Klebsiella pneumoniae* (NCBI:txid573), Gram-negative, encapsulated, non-motile, lactose-fermenting, oxidase-negative rod of the *Enterobacteriaceae*.

Environmental reservoirs & transmission. The **gastrointestinal tract is the principal reservoir**; the organism also colonizes the oropharynx and skin, and persists on hospital surfaces and medical devices, enabling patient-to-patient nosocomial transmission via healthcare-worker hands and equipment. Genomic clustering of neonatal isolates across 27 units in 13 countries confirmed extensive nosocomial spread [PMID: 42127138](#).

Lifestyle / host-environment factors. Alcohol use and diabetes are associated with invasive hvKp disease in Asian populations; broad-spectrum antibiotic exposure is the dominant modifiable environmental driver of resistant infection (see Section 2). Timor-Leste and Ethiopian surveillance underscore that third-generation cephalosporin and emerging carbapenem resistance are widespread, particularly in LMIC settings [PMID: 42662930](#), [PMID: 42594141](#).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating event → clinical manifestation)

1. **Gut/mucosal colonization** by *K. pneumoniae* (often a resistant or hypervirulent clone selected by prior antibiotics) **establishes a reservoir** in the GI tract [PMID: 32576652](#), [PMID: 42127138](#).
2. **Barrier breach** — via micro-aspiration into the lung, a urinary/vascular catheter, surgery, or translocation across a compromised gut epithelium — **allows the organism to reach a normally sterile site** (*inferred from reservoir + device risk-factor data*).
3. At the tissue site, the **polysaccharide capsule and hypermucoviscosity (hvKp: K1/K2, *rmpADC*) resist phagocytosis and complement**, while **siderophores (aerobactin, salmochelin, yersiniabactin) scavenge host iron, enabling bacterial survival and replication** [PMID: 42021129](#).
4. Bacterial ligands (LPS) **engage TLR4 on epithelium/macrophages**, which **triggers NF-κB signaling and cytokine/chemokine release (KC, MIP-2, LIX)**, recruiting neutrophils and monocytes/macrophages to the lung [PMID: 36018281](#).
5. In parallel, the **NLRC4 inflammasome senses the organism and drives IL-1β (and IL-17A) production**, which is **essential for neutrophil-mediated clearance and host survival** [PMID: 22547706](#).

6. **Branch A — effective immunity:** neutrophil/macrophage phagocytosis and inflammasome/IL-1 β signaling **clear the infection → recovery.**
7. **Branch B — failed immunity or antibiotic-resistant organism:** the bacterium **evades clearance and disseminates into blood → bacteremia/sepsis**, activating coagulation (elevated D-dimer) and causing **organ dysfunction and death (30–50% in resistant BSI)** [PMID: 42518856](#), [PMID: 42734011](#).
8. **Branch C — hypervirulent dissemination:** hvKp seeds the **liver (pyogenic abscess)** and metastasizes to **eye (endophthalmitis), meninges, lung**, producing invasive syndrome even in immunocompetent hosts [PMID: 41356976](#).

Detail by category

- **Molecular pathways:** TLR4 → MyD88 → NF- κ B (pro-inflammatory transcription); NLRC4 inflammasome → caspase-1 → IL-1 β /IL-18 maturation.
- **Cellular processes:** acute inflammation, neutrophil chemotaxis and phagocytosis, macrophage activation, pyroptosis (inflammasome-linked cell death).
- **Immune involvement:** innate immunity is decisive; NLRC4-derived **IL-1 β specifically** (not IL-18 or IL-17A) partially rescued survival and neutrophil accumulation in NLRC4 $-/-$ mice [PMID: 22547706](#). TLR4-agonist (3D-PHAD) pretreatment increased phagocytic innate cells, reduced lung CFU, and improved survival [PMID: 36018281](#).
- **Bacterial "protein dysfunction":** capsule and siderophore systems are gain-of-virulence; carbapenemases are enzymatic drug-inactivation.
- **Tissue damage:** neutrophilic inflammation, abscess/necrosis (liver in hvKp), and sepsis-associated microvascular injury/coagulopathy.

Suggested ontology terms. GO:0006954 (inflammatory response); GO:0002224/GO:0038123 (toll-like receptor signaling); GO:0072559 (NLRP/NLRC4-type inflammasome complex — verify exact ID); GO:0006935 (chemotaxis); GO:0006909 (phagocytosis). **Cell types (CL):** CL:0000775 (neutrophil), CL:0000235 (macrophage), CL:0000583 (alveolar macrophage), CL:0000066 (epithelial cell). **Anatomy (UBERON):** UBERON:0002048 (lung), UBERON:0002107 (liver), UBERON:0000178 (blood).

7. Anatomical Structures Affected

- **Primary organs:** **lung** (UBERON:0002048) — pneumonia; **blood** (UBERON:0000178) — bacteremia; **urinary tract/bladder** (UBERON:0001255) — UTI.
 - **hvKp target organs:** **liver** (UBERON:0002107) — pyogenic abscess; with metastasis to **eye** (UBERON:0000970, endophthalmitis) and **meninges/CNS** (UBERON:0002360, meningitis).
 - **Body systems:** respiratory, cardiovascular (sepsis), hepatobiliary, urinary, and (neonatal) systemic.
 - **Tissues/cells:** respiratory and urinary **epithelium** (adhesion via type 1/type 3 fimbriae *fimH/mrkD*); recruited **neutrophils** (CL:0000775) and **macrophages/alveolar macrophages** (CL:0000235/CL:0000583).
 - **Subcellular compartments (host innate response):** cytosolic inflammasome complex, plasma-membrane TLR4 receptor; (bacterial) outer-membrane capsule and LPS.
 - **Lateralization:** pneumonia may be lobar (classically upper-lobe in Friedländer pneumonia) or bilateral; liver abscess is often solitary right-lobe.
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8. Temporal Development

- **Onset:** typically **acute** (hours–days). Neonatal sepsis presents early/late neonatal period; nosocomial pneumonia after device exposure/ICU stay.
 - **Progression:** can be **rapid**, progressing to sepsis and septic shock, particularly with resistant organisms or delayed effective therapy. hvKp abscess may be subacute but disseminate.
 - **Course:** acute and, with treatment, often self-limited to the treatment course; complicated/resistant infections have prolonged courses. Not relapsing-remitting or chronic in the Mendelian sense.
 - **Critical intervention window:** early appropriate (susceptibility-matched) antimicrobial therapy is the key modifiable determinant of survival; delays worsen outcome. Faster microbiological clearance with ceftazidime-avibactam (HR 2.475; 95% CI 1.493–4.102) underscores the value of rapid, correct therapy [PMID: 42708480](#).
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9. Inheritance and Population

Inheritance. Not applicable — infectious, non-heritable. No AD/AR/X-linked pattern; no penetrance/expressivity/ anticipation/founder-effect/consanguinity considerations for the human host.

Epidemiology. - *K. pneumoniae* is the **leading cause of neonatal sepsis in LMICs** in Africa and Asia; in a genomic study of 1,523 isolates from 27 neonatal units across 13 countries, an estimated **68.0% (1,035/1,523) of neonatal infections were part of nosocomial transmission clusters**, with $\geq 57.7\%$ (879) acquired via nosocomial transmission [PMID: 42127138](#). - In ICU/ECMO cohorts, *K. pneumoniae* is repeatedly among the top VAP/HAP pathogens; VAP reached 31.1 cases/1,000 ECMO-days [PMID: 42685911](#). - CRKP bloodstream infection incidence was $\sim 2.2/100,000$ patient-days in one Chinese study [PMID: 42734011](#).

Demographics. BSI cohorts show **male predominance ($\sim 67\text{--}70\%$)** and **older age (mean $\approx 64\text{--}66$ years)** [PMID: 42518856](#), [PMID: 42563676](#). Geographic variation in carbapenemase type is pronounced: KPC-dominant in East Asia/parts of the Americas, OXA-48-dominant in the Middle East/Mediterranean, NDM widespread in South Asia. hvKp invasive liver-abscess syndrome is most reported in East/Southeast Asia.

10. Diagnostics

Culture + identification. The reference approach is **culture with MALDI-TOF MS identification** plus **molecular/phenotypic resistance detection**. MALDI-TOF correctly identified **174/174** *K. pneumoniae* isolates with log score >2.0 , 100% concordant with reference systems [PMID: 27239799](#).

Rapid molecular panels. BioFire FilmArray Blood Culture Identification from positive blood cultures showed **94% concordance** with MALDI Biotyper and detects *blaKPC* [PMID: 32305272](#).

Resistance detection. PCR/WGS identify carbapenemase genes (*blaKPC*, *blaNDM*, *blaOXA-48*, *blaVIM*); phenotypic tests include the modified Carbapenem Inactivation Method (mCIM). The **MALDIxin test** detects lipid A modifications (L-Ara4N/pEtN) underlying **colistin resistance in <30 min** and distinguishes chromosomal vs MCR-mediated resistance [PMID: 31580426](#). Strain typing (rep-PCR, WGS/MLST) supports outbreak investigation [PMID: 21568752](#).

Biomarkers / labs. Leukocytosis, elevated CRP/procalcitonin; **elevated D-dimer** is an independent prognostic marker in BSI (see Section 11). Imaging: chest radiograph/CT for pneumonia (classic bulging-fissure lobar consolidation); abdominal CT/ultrasound for hvKp liver abscess.

Differential diagnosis. Other Gram-negative pneumonias/BSI (*E. coli*, *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, *Serratia*), *S. aureus* pneumonia, and other causes of liver abscess (*E. coli*, anaerobes, amebic abscess).

11. Outcome / Prognosis

Mortality is high, especially for resistant bloodstream infection.

Outcome metric	Value	Source
In-hospital mortality, KP BSI (n=222)	47.7% (106/222)	PMID: 42518856
28-day mortality, BSI in acute-on-chronic liver failure	49.1%	PMID: 42717128
Donor-derived / hospital-acquired pneumonia in lung transplant	50%	PMID: 42781399

Prognostic factors. Independent predictors and correlates of death include **elevated D-dimer** (adjusted HR 1.04 per unit, 95% CI 1.03–1.07; high vs low group HR 2.14, 95% CI 1.27–3.59) [PMID: 42518856](#); **capsular type K47/K64 and KPC-plasmid features** in CRKP bacteraemia [PMID: 42216056](#); disease severity scores (MELD, Pitt bacteremia score, APACHE II); and **choice of therapy** — non-polymyxin regimens achieved higher clinical cure than polymyxin-based therapy in CRKP bacteraemia (58.4% vs 15.4%; $p=0.02$) [PMID: 42455851](#).

Complications. Septic shock, multi-organ failure, DIC/coagulopathy; for hvKp, metastatic endophthalmitis, meningitis, and septic emboli. Recovery is achievable with early appropriate therapy but is compromised by resistance and host frailty.

12. Treatment

Susceptibility-guided antibiotic therapy is the cornerstone. For **non-metallo- β -lactamase CRE**, IDSA 2022 guidance and the German MDRO guideline recommend **ceftazidime-avibactam, meropenem-vaborbactam, and imipenem-relebactam** as first-choice agents; **metallo- β -lactamase (e.g., NDM) production must be detected or excluded** because it changes therapy to **cefiderocol or ceftazidime-avibactam plus aztreonam** [PMID: 35439291](#), [PMID: 42661422](#).

Scenario	Preferred therapy	NCIT-type intervention
Susceptible <i>K. pneumoniae</i>	β -lactam per susceptibility (e.g., ceftriaxone)	Cephalosporin therapy
ESBL producer	Carbapenem	Carbapenem therapy
CRE (non-MBL: KPC/OXA-48)	Ceftazidime-avibactam, meropenem-vaborbactam, imipenem-relebactam	β -lactam/ β -lactamase-inhibitor therapy
MBL (NDM/VIM)	Cefiderocol, or ceftazidime-avibactam + aztreonam	Siderophore-cephalosporin therapy
Adjunct / renal-sparing	IV fosfomycin-containing regimens	Fosfomycin therapy

Ceftazidime-avibactam produced faster microbiological clearance than alternatives [PMID: 42708480](#). **IV fosfomycin** combination regimens achieved 79.4% clinical success and were associated with **reduced acute kidney injury** versus colistin-heavy regimens [PMID: 42599357](#), [PMID: 42482780](#). **Colistin/polymyxins** are now later-line owing to nephrotoxicity and inferior cure rates [PMID: 42455851](#).

Source control (drainage of liver abscess/empyema, removal of infected catheters/lines) is essential, particularly for hvKp.

Adjunctive/experimental. In a murine model, **atorvastatin added to imipenem** provided additional benefit in Gram-negative pneumonia [PMID: 29463546](#); immunomodulation of the ICOS pathway influenced airway-pathogen pathogenesis [PMID: 29378030](#). TLR4-agonist innate stimulation was protective in mice [PMID: 36018281](#). These are pre-clinical.

13. Prevention

Primary prevention. Antibiotic **stewardship** (limiting carbapenem/glycopeptide/quinolone pressure — the strongest modifiable risk factors, Section 2) and **infection prevention and control**: hand hygiene, contact precautions, device-care bundles (ventilator, central-line, urinary-catheter), and environmental cleaning to interrupt nosocomial transmission [PMID: 42127138](#), [PMID: 42685911](#).

Secondary prevention. Active surveillance cultures / rectal screening for CRKP carriage in high-risk units; early detection of carbapenemase producers enables cohorting and rapid appropriate therapy.

Vaccines (investigational). No licensed *Klebsiella* vaccine exists. Candidates target capsular polysaccharides, O-antigen (LPS), and the **MrkA** type-3 fimbrial subunit. A **pVAX1-MrkA DNA vaccine protected 80% of mice against sepsis (p=0.0373)**, identifying MrkA as a lead antigen [PMID: 42634491](#); reverse-vaccinology antigen discovery is ongoing [PMID: 42511741](#), [PMID: 42453653](#).

Microbiome-based prevention (emerging). Because the gut is the reservoir, strategies preserving/restoring a colonization-resistant microbiome are under investigation [PMID: 41009869](#).

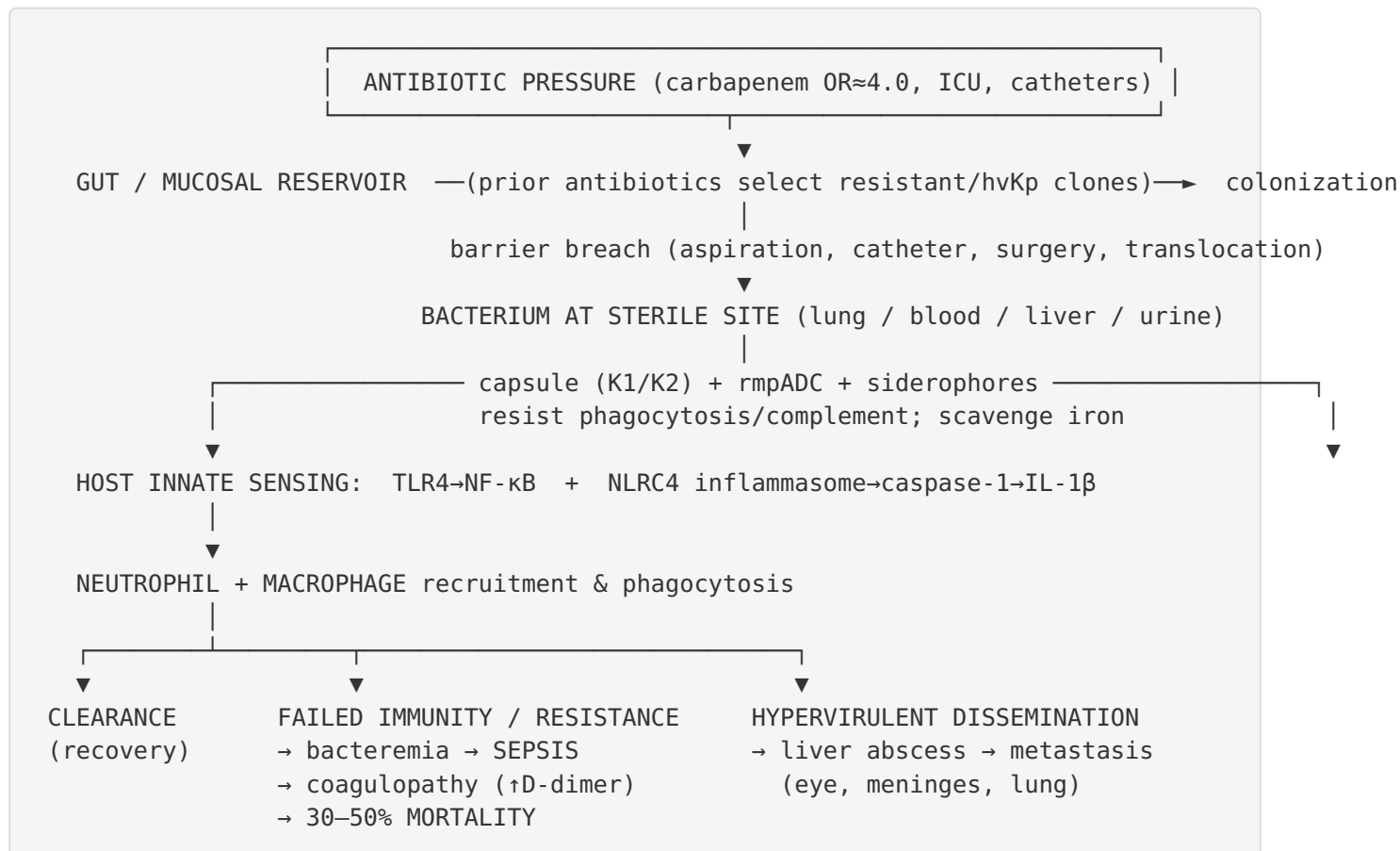
14. Other Species / Natural Disease

- **Taxonomy of pathogen:** *Klebsiella pneumoniae* (NCBI:txid573); related species *K. oxytoca*, *K. variicola*, *K. aerogenes*.
 - **Host range:** broad — *K. pneumoniae* naturally infects humans and many animals (cattle mastitis, equine and companion-animal infections, primates, wildlife) and colonizes environmental niches (water, soil, plants). It is an important **One Health** organism, with resistance genes shared across human, animal, agricultural, and environmental compartments [PMID: 42568440](#).
 - **Zoonotic/cross-species transmission:** plausible via the environment and food chain; direct zoonosis is less defined than the environmental resistance-gene flow. No single human ortholog gene is relevant (infectious disease).
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15. Model Organisms

- **Mouse (murine) models** are the primary system for pathogenesis and immunity: pulmonary infection models demonstrated the essential role of **NLRC4/IL-1 β** [PMID: 22547706](#) and **TLR4-agonist protection** [PMID: 36018281](#); sepsis models validated the **MrkA DNA vaccine** [PMID: 42634491](#) and **atorvastatin+imipenem** adjunctive therapy [PMID: 29463546](#).
 - **Genetic host models:** knockout mice (e.g., *Nlrc4*^{-/-}, *Icos*^{-/-}) dissect host-defense pathways [PMID: 29378030](#).
 - **Bacterial genetic tools:** CRISPR/Cas9 plasmid curing established causality of the hvKp virulence plasmid [PMID: 42021129](#).
 - **Phenotype recapitulation:** murine pneumonia/sepsis models reproduce neutrophilic lung inflammation, bacteremia, and (for hvKp) dissemination, and predict vaccine/therapy efficacy. **Limitations:** mouse innate immunity and microbiome differ from humans; models may not capture chronic colonization dynamics, human device-associated disease, or the full breadth of resistance-plasmid ecology. General animal-model reviews are available [PMID: 38559342](#).
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Mechanistic Model / Interpretation



The unifying insight is that **two epidemiologically distinct pathotypes share a common virulence toolkit** (capsule + siderophores) and a **common gut reservoir**, but differ in their principal threat: cKp weaponizes **antibiotic resistance** against debilitated hosts, whereas hvKp weaponizes **capsule-based hypervirulence** against healthy hosts. Their **convergence on mobile plasmids** (resistant *and* hypervirulent, e.g., ST11 KPC-2 hvKp) is the field's chief emerging concern.

Evidence Base

PMID	Contribution	Role
42127138	Neonatal sepsis, 68% nosocomial clusters across 13 countries	Establishes burden & nosocomial transmission
42734011	KPC 88.9%, NDM 7.4% in CRKP BSI; virulence gene profile	Resistance & virulence genetics
42439974	OXA-48 predominant (50.4%) in Saudi CRE	Geographic resistance variation
42021129	ST111 hvKp; 181 kb virulence plasmid; CRISPR curing	hvKp causal genetics
41356976	53-case systematic review of hvKp liver abscess	hvKp clinical phenotype
22547706	NLRC4/IL-1 β essential for pulmonary defense	Host mechanism
36018281	TLR4 agonist protective in mice	Host mechanism / therapy lead
42518856	47.7% BSI mortality; D-dimer HR 1.04	Prognosis/biomarker
42455851	Non-polymyxin cure 58.4% vs 15.4%	Treatment outcome
42216056	K47/K64 + KPC-plasmid features & mortality	Prognostic genetics
35439291	IDSA 2022 CRE treatment guidance	Treatment standard
42661422	German MDRO guideline first-line agents	Treatment standard
42708480	CAZ-AVI faster clearance HR 2.475	Treatment efficacy
31525540	Meta-analysis of CRKP risk factors	Etiology/risk
27239799	MALDI-TOF 174/174 identification	Diagnostics
32305272	FilmArray BCID 94% concordance, blaKPC	Diagnostics
31580426	MALDIxin colistin-resistance test	Diagnostics
32576652	Stool reservoir of ST11 KPC-2 hvKp	Reservoir/convergence
42634491	MrkA DNA vaccine 80% protection	Prevention
42178970	ESKAPE priority designation	Public-health framing

Limitations and Knowledge Gaps

- **Template mismatch.** This template is designed for Mendelian/genetic diseases; sections on human causal genes, inheritance, penetrance, chromosomal abnormalities, and germline variants are **not applicable** to an infectious disease and were reframed to bacterial genetics.
 - **Human host-susceptibility genetics** are underexplored — no replicated GWAS loci were identified; the protective role of TLR4/NLRC4 is inferred from mouse models, not human association data.
 - **Geographic bias.** Resistance and hvKp data skew toward East Asia, the Middle East, and select LMIC surveillance sites; global prevalence/incidence per 100,000 is not uniformly quantified.
 - **Prognostic evidence** derives largely from single-center retrospective cohorts (risk of confounding by indication for therapy choices).
 - **Vaccine and adjunctive-therapy data** are pre-clinical (murine); none is licensed or validated in humans.
 - **Convergent resistant-hypervirulent clones** (e.g., ST11 KPC-2 hvKp) are documented but their population-level frequency and outcome impact remain incompletely quantified.
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Proposed Follow-up Experiments / Actions

1. **Human host-genetics study:** targeted or genome-wide association analysis of *TLR4*, *NLRC4*/*NLRP*, *IL1B*, and complement variants against severe *K. pneumoniae* sepsis to test the inferred protective pathways in humans.
2. **Prospective, multicenter trial** comparing ceftazidime-avibactam-based vs fosfomycin-combination vs polymyxin-based regimens for CRKP BSI, powered for mortality and nephrotoxicity, stratified by carbapenemase class.
3. **Reservoir-interruption trial:** evaluate microbiome-preserving stewardship or decolonization strategies to reduce endogenous CRKP/hvKp infection in ICU cohorts.
4. **Convergent-clone surveillance:** systematic WGS monitoring for resistant-plus-hypervirulent plasmids (rmpADC + carbapenemase) to quantify emergence and outcomes.

5. **Advance MrkA/capsule/O-antigen vaccine candidates** toward first-in-human immunogenicity trials, prioritizing high-risk populations (neonates in LMICs, ICU/transplant patients).
 6. **Validate D-dimer and capsular-type (K47/K64) prognostic markers** in prospective cohorts and integrate into a clinical risk-stratification tool.
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Report generated from a 5-iteration autonomous literature synthesis (10 confirmed findings, 46 papers). Evidence types span human clinical cohorts/meta-analyses, genomic epidemiology, murine model studies, and in-vitro diagnostics. All quantitative claims are attributed to the cited PMIDs.

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